MUCSTA
MANGALORE UNIVERSITY COMP. SC. TEACHERS' ASSOCIATION

Question Bank FOR

Data Base Management System

III Semester **BCA**

2025-2026
SEP Syllabus

MANGALORE UNIVERISTY COMPUTER SCIENCE TEACHERS' ASSOCIATION

Bachelor of Computer Application (B.C.A) Programme
III Semester (SEP)
DATABASE MANAGEMENT SYSTEM
QUESTION BANK

UNIT-I

TWO MARKS

1. What is data and database?
2. Define DBMS.
3. Define query and transaction.
4. List the characteristics of Database approach.
5. Give the disadvantages of file processing system.
6. Define Metadata. Give an example
7. What do you mean by program data independence and program-operation independence?
8. Write about any four advantages of DBMS?
9. Who is DBA? What are the responsibilities of a DBA?
10. Who are database designers? What are their responsibilities?
11. Define Entity, Attribute and relationship
12. What is database schema and Schema diagram?
13. What is the difference between logical and physical independence?
14. What is Data Definition language?
15. What is Data Manipulation language?
16. Define a) meta-data b) data dictionary
17. List various types of attributes.
18. What is data model and list the types of data model used?
19. Define Single-Valued versus Multivalued Attributes, give example
20. Define Stored versus Derived Attributes, give example.
21. Define Composite versus Simple (Atomic) Attributes, give example
22. Define the terms i) Entity set ii) Relationship set
23. Define the terms i) Key attribute ii) Value set(Domain)
24. What do you mean by relationship type and relationship instances?
25. Define “degree of relationship type”.
26. What does the cardinality ratio specify? Which are the possible cardinality ratios for binary relationship types
27. Define the terms i) Total participation ii) Partial Participation
28. Define weak entity types?

THREE OR MORE MARKS

1. What are the characteristics of DBMS? Explain
2. Explain any 5 advantages of DBMS.
3. Explain the different categories of End Users in DBMS?
4. Write a short note on i) DBA ii) Database Designers
5. What is data Model? Explain different categories of data models?
6. Explain three schema architecture of database with a neat diagram
7. What is data Independence? Explain their types
8. What are the different database languages? Explain with example
9. What are the User-friendly interfaces provided by DBMS? Explain any four.
10. Explain the classification of Database Management Systems
11. Explain different types of attributes with example.
12. Write a short note on Entity types and Entity sets.
13. Explain the degree of a relationship type .Give an example for binary and a ternary relationship.
14. Define cardinality ratio? Explain its types
15. Explain different types of symbols used in ER Diagram.
16. What is participation constraint? Explain types of participation.
17. Explain binary and ternary degree of a relationship type.
18. Define the following terms: a. Single-valued and multi-valued attributes b. Primary key and foreign key
19. Write a note on Weak entity types.
20. Explain types of participation constraints.
21. Draw an ER Diagram for company considering the following constraints −
    - In a company, an employee works on many projects which are controlled by one department.
    - One employee supervises many employees.
    - An employee has one or more dependents.
    - One employee manages one department.

UNIT-II

TWO MARKS

1. What is the degree of a relation? What is the degree of the following relation? STUDENT (Name, SSn, Home_phone, Address, Office_phone, Age, Gpa)
2. What is relational model?
3. What is relation schema? List the characteristics of Relations
4. Define a) tuple b) Attribute c) Domain

5. List any two categories of relational model constraints.
6. Define super key with an example
7. What is candidate key? Give an example
8. What is entity integrity constraint and referential entity constraint?
9. Mention the properties of key in a relational schema
10. What is entity integrity constraint and referential integrity constraint?
11. Mention different update operation in a Relational model.
12. What is the function of Insert, update and delete functions?
13. Mention any two update anomalies.
14. State interpretations of NULL values
15. Define functional dependency.
16. What do mean by “normalization of data”?
17. What is 1NF,2NF?
18. What is 3NF?
19. What is BCNF?
20. Differentiate between fully and partial functional dependencies?

THREE OR MORE MARKS

1. Explain different relational model concepts with examples.
2. Explain the term Domain, tuple and relation schema with example.
3. Explain the characteristics of Relations.
4. Explain the categories of constraints
5. Write a note on key constraints
6. Explain Entity integrity and referential integrity constraints
7. Explain different update operations in a Relational model.
8. Explain different update anomalies with examples?
9. Explain functional dependencies with example
10. Write a note on “normalization of relations”
11. Explain 1NF with an example.
12. Explain 2NF with an example.
13. Explain 3NF with an example.
14. Explain BCNF with an example.
15. Find closure set of functional dependency
    i) R(ABCDEFG)
       A-> B
       BC->DE
       AEG->G
       Find Closure set of $(AC)^+$
    ii) R(ABCDE)
       A→ BC

CD→ E
B→ D
E→ A
Find Closure set of $(B)^+$

iii)R(ABCDEF)
AB→ C
BC→ AD
D→ E
CD→B
Find Closure set of $(AB)^+$

iv)R(ABCDEFGH)
A→ BC
CD→ E
E→C
D→ AEH
ABH→ BD
DH→ BC
Find Closure set of $(BCD)^+$

16. Find number of candidate key
    i) R(ABCD)
    A->B
    B->C
    C->A
    ii)R(ABCD)
    AB->CD
    D->A
17. Given a relation R(P, Q, R, S, T) and Functional Dependency set FD = { PQ → R, S → T }, determine whether the given R is in 2NF? If not convert it into 2 NF.
18. Given a relation R(P, Q, R, S, T, U, V, W, X, Y) and Functional Dependency set FD = { PQ → R, PS → VW, QS → TU, P → X, W → Y }, determine whether the given R is in 2NF? If not convert it into 2 NF.

UNIT-III

TWO MARKS

1. Write the syntax of create table command?
2. How do you load the data to the table? Write its syntax.
3. Write the syntax of update command.
4. Write the syntax of alter command.
5. Write the syntax of delete command.
6. Write the syntax of SELECT command.
7. How can you RENAME the tables.
8. List any four **aggregate** functions used in SQL.
9. List any four **date** functions used in SQL.
10. List any four types of data constraints.

11. What do you mean by NOT NULL constraint?
12. Why do we use check constraint?
13. How do you we ALTER command to drop the constraints?
14. Mention the command used to modify the column definitions? Give example.
15. What do you mean by DEFAULT clause in SQL?
16. How do you sort the records of a table in SQL? Give one example.
17. What is JOIN operation? Give one example.
18. List types of join used in SQL?
19. What is dual table?
20. What do you mean by GROUP BY clause? Give one example.
21. What is sub query?
22. What is privilege? Write the syntax of granting the privilege to the user.
23. Write the syntax of REVOKE statement? Give one example.
24. Why do we use LIKE operator in SELECT statement? Give one example.
25. What is the importance of primary key in a table?
26. How to create primary constraint?
27. Differentiate primary key constraint and unique key constraint.
28. What do mean by value "NULL" in SQL?

THREE OR MORE MARKS

1. How do you filter the data from the table? Explain the command use, with example.
2. Explain any 5 aggregate functions with syntax and example in Oracle.
3. What is data constraint? List and explain any two data constraint with syntax and example.
4. What do you mean by foreign key constraint? What are the features of foreign key
5. Explain ORDER BY clause with example?
6. Create the table described below: Table name: CLIENT_MASTER

Column Name	Data type	Size	Attributes
CLIENT_NO	Varchar2	6	Primary key/first letter must start with C, NOT NULL
NAME	Varchar2	20	
ADDRESS1	Varchar2	30	
ADDRESS2	Varchar2	30	
CITY	Varchar2	15	
PINCODE	Number	8	NOT NULL
STATE	Varchar2	15	NOT NULL
BALDUE	Number	10,2	NOT NULL

Consider the table CLIENT_MASTER given , write the following queries:

a) Find out the names of the clients with CLIENT_NO='C00001'
b) Change the city of CLIENT_NO 'C00005' to 'Bangalore'
c) Retrieve the list of names, city and state of all the clients.

d) Delete from CLIENT_MASTER where the column STATE holds the value 'TAMILNADU'
e) Add a column called 'TELEPHONE' of data type 'NUMBER' and size=10 to the CLIENT_MASTER table.
f) Destroy the table CLIENT_MASTER along with its data.
g) Change the name of the table CLIENT_MASTER to C_M.

7. List and explain five clauses used in SELECT statement.
8. Explain the types of joins used in sql with example
9. List and explain any 6 data types used in oracle.
10. How to create a table from another table? Explain with syntax and example.
11. How do you insert data into a table from another table? Explain with syntax and example
12. How do you create a foreign key constraint in a table definition? Explain with example.
13. What do you mean by primary key constraint/ how can we create primary keys in a table? Explain with syntax and example.
14. Explain CHECK constraint with syntax and example.
15. What are the arithmetic and logical operators used in SQL? Explain with example.
16. Explain 'BETWEEN' and 'LIKE' operators with example.
17. Define these operators: EXISTS,NOT EXISTS,IN, ALL, DISTINCT with example
18. Explain UNION and INTERSECT clause used in SQL, with example.
19. Consider the tables given below: PRODUCT_DETAIL(PRODUCT_NO, PRODUCT_NAME,DESRIPTION, QTYACLB, PRICE, PROFITIN%) PURCHASED_DETAIL (CUSTNAME, PRODUCTNO, QTYSOLD). Write the following queries:
    a) Print the description and qtysold for each product.
    b) Find out the products, which have been sold to 'Ivan'
    c) Find productno and description of nonmoving products(i.e. products not being sold).

UNIT IV

TWO MARKS

1. Give two advantages of PL/SQL over SQL.
2. Give the structure of the PL/SQL program.
3. What is literal in PL/SQL? Mention two types of literals.
4. What is oracle transaction?
5. What is the purpose of COMMIT command?
6. Give the General format of while Loop construct in PL/SQL
7. Give the syntax of for loop construct in PL/SQL.
8. What is use of ROLLBAK command?
9. What is the use of SAVE POINT in oracle transaction?
10. Give the syntax of SELECT statement in PL/SQL.
11. What is cursor?
12. Differentiate implicit and explicit cursors.

13. List four cursor attributes.
14. Mention types of implicit cursor attributes.
15. Mention types of explicit cursor attributes.
16. What is the command used to read the cursor data? Give syntax.
17. Differentiate between cursor with parameters and without parameters.
18. What is an exception? How it is handled in PL/SQL?
19. What is difference between function and stored procedure?
20. Give the syntax of defining stored procedure in Pl/SQL.
21. Give the syntax of defining function in Pl/SQL.
22. Differentiate between IN and OUT parameters of procedure/function.
23. Mention two advantages of using PACKAGE in Oracle.
24. What are the components of oracle package?
25. Give the general format of PACKAGE specification.
26. Give the General format of PACKAGE body.
27. What is a trigger? What are three parts of trigger?

THREE OR MORE MARKS

1. Define PL/SQL. Explain the advantages of PL/SQL.
2. Explain the structure PL/SQL program with example
3. What are the different types of literals ? Explain with example
4. Explain the syntax and usage of the following with example :
   a. While loop b) for loop c) if .. endif
5. Write a note on labelled loops.
6. Explain the commit and rollback commands used in oracle transaction with example
7. What is a cursor? explain the different types of cursor
8. Explain explicit cursor attributes with example for each.
9. Explain use of parameterised cursors with example
10. Explain with syntax and example, how to create a cursor and read the records from a cursor.
11. Explain all the attributes of a cursor
12. Write a note on error handling in PL/SQL
13. Explain any three system exceptions of PL/SQL
14. What is a procedure? What are the advantages of procedures?
15. What is a functions? What are the advantages of functions?
16. Explain with example how to create a procedure and use it . Explain the keyword and parameters used for creating procedures
17. Explain with example how to create a function and use it .Explain the keyword and parameters used for creating function.
18. Explain different data types supported by PL/SQL.
19. Explain various commands associated with oracle transaction with example.
20. Explain various forms of if..endif structures with syntax and example
21. Explain various looping statements of PL/SQL with syntax.
22. Explain various commands/statements required to open , process and close

the explicit cursor with syntax and example.

23. With syntax and example explain how a stored procedure is created and executed in PL/SQL ?
24. With syntax and example explain how a function is created and executed in PL/SQL?
25. With syntax and example explain how a package is created and executed in PL/SQL?
26. With example explain how user defined exception is handled in a PL/SQL program.
27. What do you mean by an exception? Explain any five pre-determined exceptions
28. How to create a trigger? Explain with syntax and example.

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